

Project 3 — Individual Contribution Report

Name: Blake Simpson

Course: CS 5542 — Big Data Analytics and Applications

Project: Integrated System Report and Final System Enhancement

Date: March 30, 2026

Personal Contributions

1. Project Scaffolding and Initial Architecture

Set up the initial repository structure, including stub files for the FastAPI backend, Streamlit frontend, Snowflake schema, data ingestion script, retrieval module, and evaluation scaffold. This established the project layout and directory conventions that the team built upon throughout the semester.

Key commits: b2f10eb (initial commit), 020a381 (project scaffolding)

2. Agentic Search Layer (Lab 6)

Implemented the AI agent reasoning loop in the FastAPI backend. The agent uses Gemini 2.5 Flash with two tool declarations (vector database search and knowledge graph search) and runs up to 5 reasoning iterations with autonomous tool selection. Added citation tracking so responses include bracketed source references tied to retrieved chunks, and added the Snowflake vector type migration script.

Key commits: 1c3eb32 (agentic search), d19b0cf (backend and frontend integration)

3. Frontend Chat UI and History

Built the Streamlit chat interface with message history display, citation rendering, and a sidebar showing past queries. Added tool call transparency so each response shows how many tool calls and agent iterations were used.

Key commits: 1d0a919 (chat history frontend), d19b0cf (frontend integration), a644713 (tool call display)

4. Metrics Dashboard (Lab 9)

Added a "Dashboard" tab to the Streamlit application with real-time system monitoring:

- Summary KPI row (total queries, average latency, average confidence, corpus size)
- Query latency over time (line chart)
- Confidence score distribution (histogram)
- Tool usage breakdown (bar chart)
- Recent queries table (last 20 queries with metadata)
- Snowflake table inventory (row counts across RAW, GRAPH, APP schemas)

Also added the `GET /metrics/history` backend endpoint and enriched the query logging to persist confidence, latency, retrieval mode, tool calls, and iteration count alongside each query.

Key commits: `a644713` (metrics dashboard), `9ff4541` (metrics dashboard PR merge)

5. Lab 8 Integration — Domain Adaptation

Integrated the team's Lab 8 domain adaptation artifacts into the main project repository so the Project 3 repo reflects all four lab enhancements. The integrated materials include the QLoRA fine-tuning notebook, GEPA prompt optimization notebook, the 50-example instruction dataset, and the evaluation notebook comparing four configurations.

Key commit: `862c94e` (lab 8 integration)

Percentage Contribution

25% of total project work.

GitHub Contribution Evidence

Commit	Date	Description
b2f10eb	2026-02-09	Initial commit — README, proposal

Commit	Date	Description
020a381	2026-02-25	Project scaffolding — backend, frontend, schema, ingestion, evaluation stubs
d19b0cf	2026-03-03	Backend and frontend integration
1c3eb32	2026-03-09	Agentic search — multi-iteration agent, tool definitions, vector migration
1d0a919	2026-03-09	Chat history frontend
a644713	2026-03-23	Metrics dashboard — frontend tabs, backend endpoint, tool call display
9ff4541	2026-03-23	Metrics dashboard PR merge
862c94e	2026-03-29	Lab 8 domain adaptation integration

Tools Used

- **Claude Code (Anthropic Claude)** — Used for planning implementations, generating code, and reviewing teammate contributions for integration. All AI-generated code was reviewed and tested before committing.

Reflection

My contributions focused on the project foundation, the agentic reasoning layer, and the frontend experience. Setting up the initial scaffolding gave the team a shared structure to work from early on. The agentic search implementation was the most technically involved piece — coordinating the LLM tool-calling loop with Snowflake-backed retrieval across multiple reasoning iterations. The metrics dashboard in Lab 9 added observability to the system, making performance trends visible for the first

time. Finally, consolidating the Lab 8 domain adaptation work into the main repository brought all four lab enhancements together in one place for Project 3.